

Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 13: Statistics (Core-Basics Edition)

CHAPTER 13: STATISTICS

A simplified, detailed guide in Hinglish
Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) DATA

Data = collected numbers/observations.

Grouped data = intervals mein bata hua (0-10, 10-20).

2) MEAN, MEDIAN, MODE (basic meaning)

Mean = average. Median = beech wali value.

Mode = sabse zyada baar aane wali value.

3) CLASS INTERVAL AUR FREQUENCY

Class interval = range (jaise 10-20).

Frequency (f_i) = us interval mein kitne items.

4) CLASS MARK (x_i)

Class mark = interval ka midpoint

= $(\text{upper} + \text{lower})/2$. (0-10 $\rightarrow$ 5)

5) SUMMATION (σ)

$\text{Sum}(f_i)$ = saari frequencies ka total.

$\text{Sum}(f_i x_i)$ = har ($f_i \times x_i$) ka total.

6) AVERAGE NIKAALNA

Average = total / count. Mean isi pe based hai.

7) CUMULATIVE FREQUENCY (cf)

cf = frequencies ko add karte jao (running total).

Median nikaalne ke liye zaroori.

NOW: ASLI CHAPTER 13 SHURU

TOPIC 1: MEAN OF GROUPED DATA

EXAM ALERT: Mean (direct method) bahut aata hai:

$$\text{Mean} = \frac{\sum(f_i x_i)}{\sum(f_i)}$$

Steps: class mark x_i nikaalo $\rightarrow f_i \cdot x_i \rightarrow$ dono ke sum
 $\rightarrow$ divide.

(Assumed-mean aur step-deviation alternate methods hain
jab numbers bade ho.)

TOPIC 2: MODE OF GROUPED DATA

EXAM ALERT: Mode formula:

$$\text{Mode} = l + \left[\frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)} \right] \times h$$

l = modal class lower limit, f_1 = uski frequency,
 f_0 = pehle wali, f_2 = baad wali, h = class size.
Modal class = sabse zyada frequency wala interval.

TOPIC 3: MEDIAN OF GROUPED DATA

EXAM ALERT: Median formula:

$$\text{Median} = l + \left[\frac{(n/2 - cf)}{f} \right] \times h$$

Pehle cumulative frequency banao. $n/2$ jis class mein
aaye = median class. cf = uske pehle wali cf ,
 f = uski frequency, h = class size.

TOPIC 4: EMPIRICAL RELATIONSHIP

$$3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$$

(Do pata ho toh teesra nikaal sakte ho.)

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples poori tarah solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - Median (full table)

(Exam favourite)

CI: 0-10,10-20,20-30,30-40 ; f: 5,8,12,5.

Median nikaalo.

cf: 5, 13, 25, 30. $n=30$, $n/2=15$.

$n/2=15$ pehli baar cf=25 (class 20-30) pe cross hota.

Median class = 20-30: $l=20$, $cf(\text{before})=13$, $f=12$, $h=10$.

$$\begin{aligned}\text{Median} &= 20 + \frac{[(15-13)/12] \times 10}{1} \\ &= 20 + (2/12) \times 10 = 20 + 1.67 = 21.67.\end{aligned}$$

Solved Example 2 - Mean (direct method)

(Exam favourite)

CI: 0-10,10-20,20-30,30-40 ; f: 5,8,12,5.

xi : 5,15,25,35. $fi \cdot xi$: 25,120,300,175.

$\sum(fi)=30$, $\sum(fi \cdot xi)=620$.

$$\text{Mean} = 620/30 = 20.67.$$

Solved Example 3 - Mode

(Exam favourite)

CI: 0-10,10-20,20-30,30-40 ; f: 3,8,12,7.

Modal class = 20-30 (max freq 12).

$l=20$, $f_1=12$, $f_0=8$, $f_2=7$, $h=10$.

$$\begin{aligned}\text{Mode} &= 20 + \frac{[(12-8)/(24-8-7)] \times 10}{1} \\ &= 20 + (4/9) \times 10 = 20 + 4.44 = 24.44.\end{aligned}$$

Solved Example 4 - Empirical relation

Mean=30, Median=28. Mode nikaalo.

$$\begin{aligned}\text{Mode} &= 3 \text{ Median} - 2 \text{ Mean} = 3(28) - 2(30) \\ &= 84 - 60 = 24.\end{aligned}$$

Solved Example 5 - Class mark

30-40 ka class mark? $\rightarrow (30+40)/2 = 35$.

Solved Example 6 - Simple mean

CI: 0-10, 10-20, 20-30 ; f: 4, 6, 10.

xi: 5, 15, 25. $f_i \cdot x_i$: 20, 90, 250 $\rightarrow$ sum=360. $\sum(f_i) = 20$.

Mean = $360/20 = 18$.

Solved Example 7 - Cumulative frequency

f: 5, 8, 12, 5. cf banao.

cf = 5, 13, 25, 30.

Solved Example 8 (EASIEST) - Modal class

f: 3, 8, 12, 7. Modal class konsa?

Max frequency 12 $\rightarrow$ modal class wahi interval.

Q AND A TIME

Q1. CORE: 20-30 ka class mark kya hoga?

Q2. Mean nikaal: CI 0-10, 10-20, 20-30 ; f 4, 6, 10.

Q3. Mode formula likho aur f_0, f_1, f_2 kya hote hain bata.

Q4. Mean=25, Median=24. Empirical se Mode nikaal.

Q5. f: 6, 10, 8, 4. Cumulative frequency banao.

Q6. Median formula mein cf ka matlab kya hai?

Q7. CORE CHECK: mean kaise nikalte hain (formula)?

SUMMARY

1. Class mark $x_i = (\text{upper} + \text{lower})/2$.

2. Mean = $\text{Sum}(f_i x_i) / \text{Sum}(f_i)$.

3. Mode = $l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$.

4. Median = $l + [(n/2 - cf)/f] \times h$.

5. $3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$.

CORE (RED page) revise: data, mean/median/mode meaning,
class mark, frequency, summation, cumulative frequency.

Generated by Kiro for Krishna1k

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